

L#02: Power, energy & circuit elements

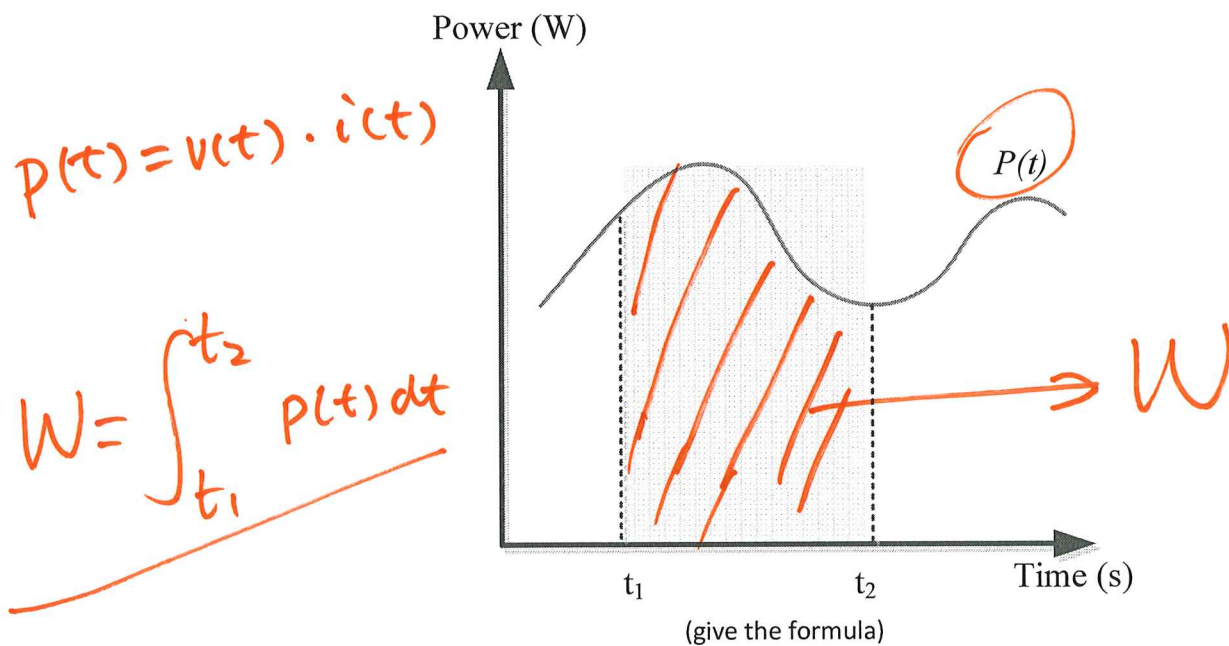
2018-02-06

Power & Energy

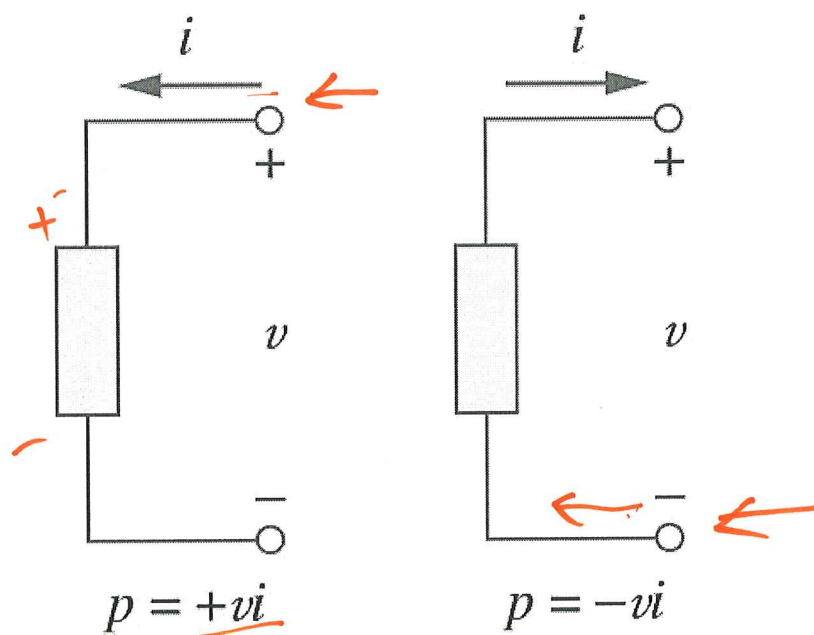
- Power: the rate at which energy is consumed or delivered.

$$W: \frac{dW \text{ (J)}}{dt \text{ (s)}} \rightarrow \text{J/s} \Rightarrow \text{Watt}$$
$$V \left[\frac{dW}{dq} \right] \cdot \left[\frac{dq}{dt} \right] I \quad P = \frac{W}{t} = \underline{V \cdot I}$$
$$W = P \times t$$

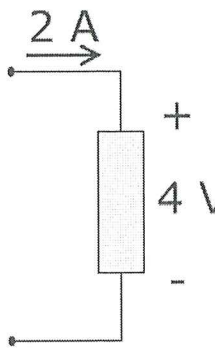
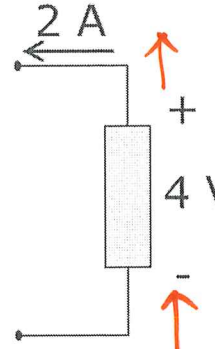
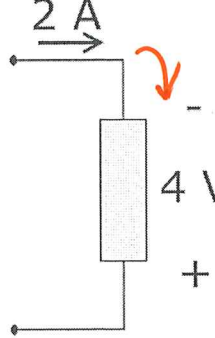
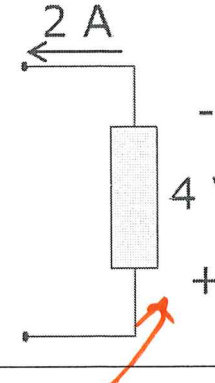
- Energy is the capacity to do work, $W \cdot s$ measured in J or Wh.
- Law of conservation: —.



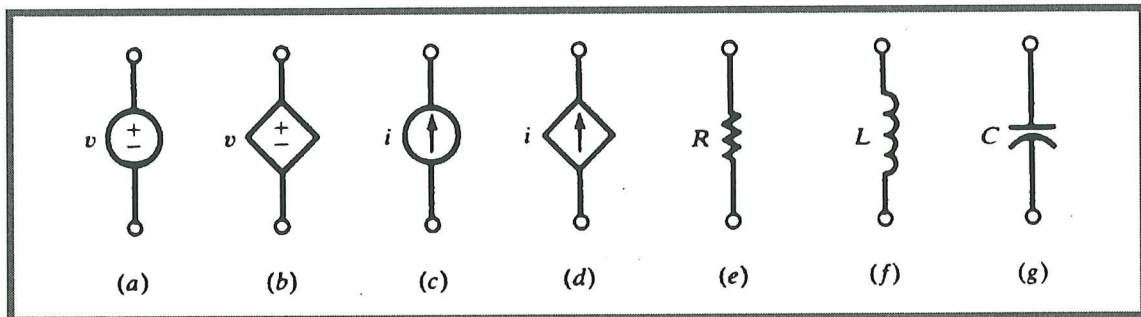
Passive sign convention



Current	Power	Energy
Enters the positive terminal	positive	consume
Enters negative terminal	negative	deliver

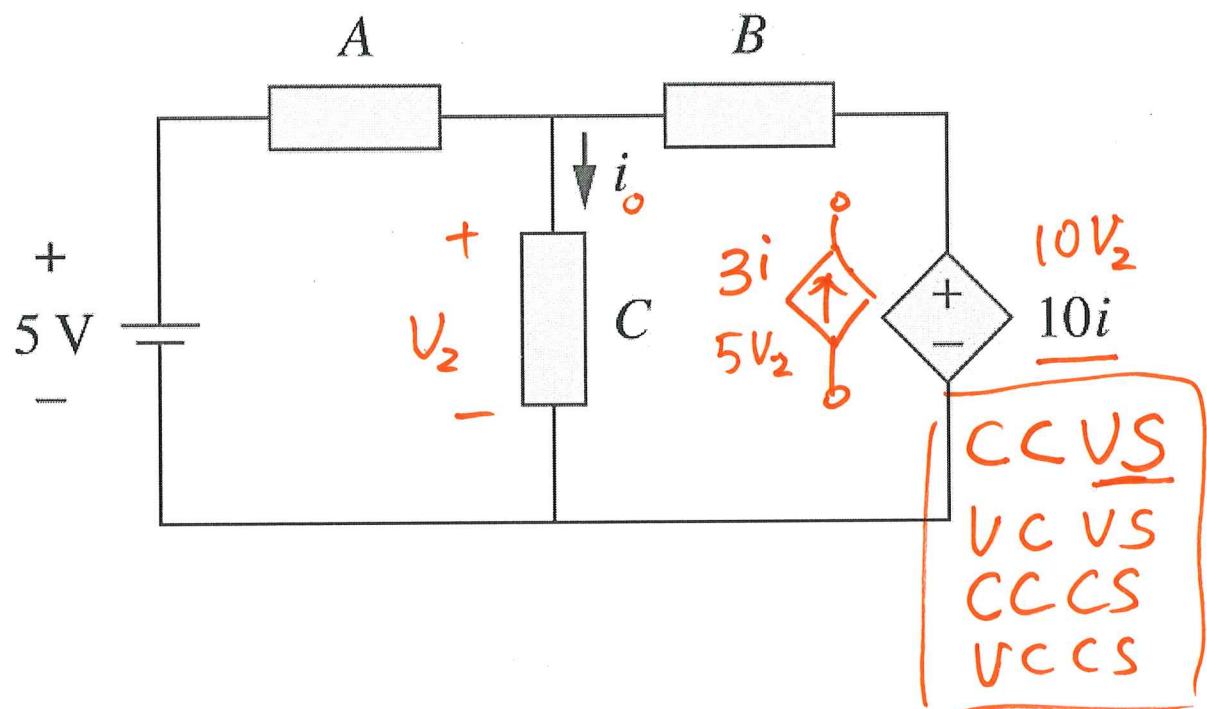
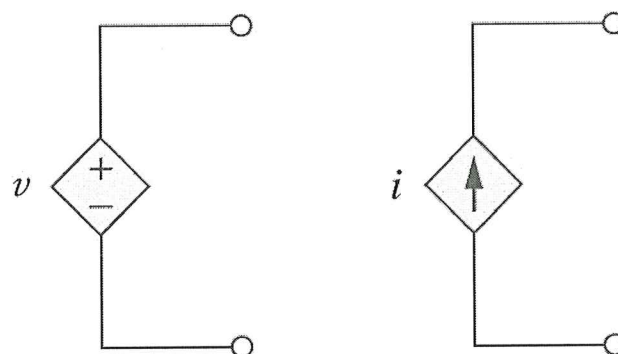
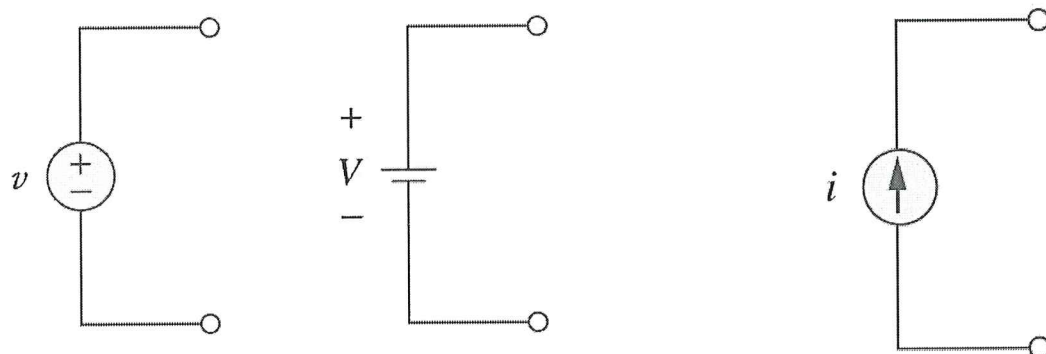
I 	$P = 2 \times 4 = 8W$
	$P = -2 \times 4 = -8W$
	$P = -8W$
	$P = 8W.$

Circuit elements



Passive	
Active	elements: - provide power - amplify signals
Independent	
Dependent	

Independent and Dependent Sources



Example #1:

$$W(t), q(t)$$

To move charge $q = 2C$ from point a to b requires -30 J . Find the voltage drop v_{ab} .

$$V_{ab} = \frac{W}{q} = \frac{-30\text{ J}}{2C} = -15\text{ V.}$$

$$V_{ab} = \frac{dw}{dq}$$

Example #2:

Find the power delivered to an element at $t = 5 \text{ ms}$ if the voltage is $v = 2i \text{ V}$ the current entering its positive terminal is: $i = 5 \cos 60\pi t \text{ A}$.

Example #3:

A stove element draws 15 A when connected to a 240 V supply. How long does it take to consume 60 kJ?

Example #4:

Calculate the power supplied or absorbed by each element.

